# LiveTimer — Feature Ideas

Features found in other classroom timers and productivity tools worth considering

---

## Already Implemented

✓ **Student Check-In / Done** — Students signal "Working on it" or "Done" from their phone. Instructor sees a live status board. *(In progress now)*

---

## Scheduled / Recurring Timers

Pre-schedule a sequence of timers that run automatically — e.g., 25-min work, 5-min break, 25-min work, 15-min break (Pomodoro). The instructor sets up the sequence once, hits start, and the timers chain together without intervention.

*Notes:* Natural extension of the library. Could add a "Sequence" concept — an ordered list of library items that auto-advance. Students see each phase with its own label/message.

## Visual / Audio Themes

Let instructors choose from a set of visual themes (colors, backgrounds, fonts) and alarm sounds. Some timers offer seasonal themes, holiday themes, or classroom-appropriate styles.

*Notes:* The alarm/warning sound system is already in place. Could add theme presets that change CSS variables. Keep it simple — 4-5 curated themes, not a full color picker.

## Persistent Timer State Across Restarts

Save the running timer state to the database so it survives server restarts and deploys. When the server comes back, it recalculates remaining time from the saved endTime and resumes.

*Notes:* Already agreed upon and in CLAUDE.md as a pending item. The keep-alive ping is the first defense; this is the safety net.

## Multiple Simultaneous Timers / Groups

Run different timers for different groups of students at the same time — e.g., Group A has 10 minutes for Task 1 while Group B has 15 minutes for Task 2. Each group sees only their timer.

*Notes:* Significant architecture change — would need room/channel concepts in Socket.io. Could start simpler: just two named timer slots. Consider whether the complexity is worth it for the use case.

## Countdown to Specific Time of Day

Instead of setting a duration, set a target wall-clock time (e.g., "Class resumes at 1:30 PM"). The timer counts down to that moment.

> *Notes:* The duration popup already has a Target Time spinner that syncs with duration. This is partially implemented — the UX is there, but it always converts to a duration before starting.

## Embedded Chat / Announcements

A simple one-way or two-way message channel between instructor and students. The instructor sends quick announcements ("Remember to save your work") that appear on student phones as a notification banner.

> *Notes:* The Message field already pushes text to students. Could extend to a scrolling notification feed or a toast-style overlay that auto-dismisses. Keep scope small — this isn't Slack.

## Analytics / Session Log

Track how timers were used in a session: which presets were loaded, actual duration vs. planned, how many students checked in, average completion times. Exportable as a report.

> *Notes:* Useful for instructors who need to document pacing for curriculum reviews. Could log events to MongoDB and offer a downloadable CSV or simple dashboard after class.

## Fullscreen / Kiosk Mode

A dedicated fullscreen mode optimized for projecting on a classroom screen — no browser chrome, no scrollbars, maximized use of screen real estate. One-click toggle.

> *Notes:* The student view already works well fullscreen. Could add a dedicated /display route or a button that triggers the Fullscreen API. OBS browser source already handles this for streaming.

## Timer Sharing / Import via Link

Generate a shareable link or QR code that contains a full timer preset (duration, label, message, options). Another instructor scans it and the preset loads into their library.

> *Notes:* Library export/import already exists as JSON. A URL-encoded single-timer share link would be more convenient for quick sharing between colleagues.

## Accessibility Improvements

Screen reader support, high-contrast mode, keyboard navigation, reduced-motion option for the progress ring animation and color transitions.

> *Notes:* Important for inclusive classrooms. ARIA labels on the timer elements, a high-contrast CSS toggle, and prefers-reduced-motion media query would cover the basics.

## Multi-Instructor Support

Allow multiple instructors to have their own independent timer sessions, each with their own library, class code, and connected students. Login / authentication system.

> *Notes:* Already noted in CLAUDE.md as a Phase 2 goal. Requires user accounts, per-user data isolation in MongoDB, and a session management layer.

## Integrations (LMS / Calendar)

Connect to learning management systems (Canvas, Blackboard) or calendar apps to auto-create timers from scheduled class activities. Pull break times from the syllabus.

> *Notes:* High effort, niche audience. Probably only worth it after multi-instructor support exists and there's real user demand. API integrations are maintenance-heavy.

---

Generated May 2026 — for planning discussion, not a commitment.